MASTER'S RESEARCH PROJECT PROPOSAL

Virtual Kidney Biopsy: Predicting Tissue Pathology from Clinical Data Using Causal Artificial Intelligence

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Provisional Project Title

Virtual Kidney Biopsy: Predicting Tissue Pathology from Clinical Data Using Causal Artificial Intelligence

Provisional Project Description

This research develops a noninvasive system to assess kidney tissue damage severity using artificial intelligence and routine medical test results, eliminating the need for surgical biopsy procedures. By analyzing patterns in blood tests, vital signs, and patient medical records from a large hospital database, the system will predict tissue damage levels and identify which medical factors contribute to kidney disease progression, enabling doctors to monitor patients continuously and choose better treatments.

1. Background to the Problem

1.1 Clinical Context

Chronic kidney disease affects 850 million people worldwide. Understanding how badly a patient's kidney tissue is damaged is essential for choosing the right treatment and predicting disease progression. Currently, doctors must perform a kidney biopsy inserting a needle into the kidney to extract tiny tissue samples to see tissue damage directly.

1.2 Limitations of Current Approach

Traditional kidney biopsies have four major problems:

1. Invasive and risky: Requires surgery with 1-2% chance of serious bleeding or infection
2. Expensive: Costs $1,000-$5,000 per procedure, limiting access for many patients
3. One time only: Cannot be repeated safely to track disease changes over time
4. Sampling limitations: Small tissue samples may miss important areas of damage

1.3 The Gap

Traditional kidney biopsies require inserting a needle into the kidney to extract tissue samples, carrying 1-2% risk of serious bleeding or infection and costing $1,000-$5,000 per procedure, yet they cannot be repeated frequently in the 850 million chronic kidney disease patients worldwide. This forces doctors to make treatment decisions without being able to monitor how the kidney tissue is responding, resulting in reactive rather than proactive care and preventing personalized treatment approaches.

2. Initial Literature Review

2.1 Current Research in Medical AI

Recent studies show that artificial intelligence can predict various medical conditions from routine test results. For example, researchers have used AI to predict heart disease risk, diabetes complications, and cancer outcomes. However, most existing kidney disease research focuses on predicting general outcomes like kidney failure, not specific tissue damage severity.

2.2 Existing Baseline

Li et al. (2017) attempted to classify kidney disease types using standard AI methods on 222 patients, achieving 62% accuracy. This establishes the baseline our research must exceed. However, their study was limited by small patient numbers and only classified disease type rather than measuring tissue damage severity.

2.3 Causal AI in Medicine

Traditional AI finds patterns (X and Y happen together) but cannot identify causes (X causes Y). New causal AI methods can identify which factors actually cause disease progression versus which factors merely correlate. This distinction is crucial for treatment planning doctors need to know which factors they can change to improve patient outcomes.

2.4 Research Gap

No existing research combines: (1) predicting kidney tissue damage without biopsy, (2) using causal AI to identify treatment targets, and (3) working with large hospital databases. This research fills that gap.

3. Research Aim

To design, develop, test, and evaluate the first noninvasive system that can assess kidney tissue damage from routine medical tests, eliminating the need for invasive biopsy procedures while enabling continuous monitoring of disease progression. By combining artificial intelligence with causal analysis methods, this research will not only predict tissue damage but also identify what causes it and help doctors choose the best treatments for each patient.

4. Research Objectives

- **Primary:** Develop AI models that predict kidney damage severity from routine blood tests and medical records without requiring surgical biopsy
- **Secondary:** Identify which medical factors cause kidney damage to worsen and which factors doctors can change to help patients
- **Tertiary:** Compare whether understanding causes improves AI predictions compared to just finding patterns

5. Research Questions

RQ1: How can artificial intelligence predict the severity of kidney tissue scarring from blood tests and medical records alone, allowing doctors to monitor patients without performing risky needle biopsies?

RQ2: Which medical factors such as blood pressure, protein in urine, or diabetes directly cause kidney scarring to worsen, and can we identify which of these factors doctors can change to slow disease progression?

RQ3: When predicting kidney tissue damage, does AI that understands cause and effect relationships perform better and provide more useful treatment guidance than AI that only finds statistical patterns in the data?

6. Operational Objectives

To accomplish the research objectives, this project will:

- Obtain access to hospital database containing 8,000+ kidney disease patients.
- Extract and clean patient data including blood tests, vital signs, and medications.
- Use text analysis to find biopsy results in medical notes.
- Build and train multiple AI prediction models (standard and advanced methods).
- Apply causal analysis methods to identify cause effect relationships.
- Validate results against medical literature and expert knowledge.
- Compare performance of different AI approaches.

7. Proposed Solution

7.1 Overall Approach

The proposed virtual kidney biopsy system, shown in Figure 1, will analyze routine medical data to predict tissue damage severity. The system uses three parallel analysis tracks that work together:

- **Track 1:** Standard AI methods (proven reliable approaches) - addresses RQ1
- **Track 2:** Advanced AI methods (captures patterns over time) - addresses RQ1
- **Track 3:** Causal analysis (finds actual causes) - addresses RQ2

7.3 Overall Study Diagram

The complete research workflow proceeds through five major phases:

Phase 1: Data Collection & Preparation - Extract patient records, find biopsy results in text notes, prepare data for analysis

Phase 2: Feature Engineering - Convert raw medical data into analyzable information (patient characteristics, test trends, calculated values)

Phase 3: Three Parallel Modeling Tracks

- **Track 1 (RQ1):** Build standard AI prediction models
- **Track 2 (RQ1):** Build advanced AI models for time patterns
- **Track 3 (RQ2):** Apply causal analysis to find causes

Phase 4: Integration - Combine all three approaches into one system

Phase 5: Validation (RQ3) - Test accuracy, compare causal vs non causal approaches, validate with medical experts

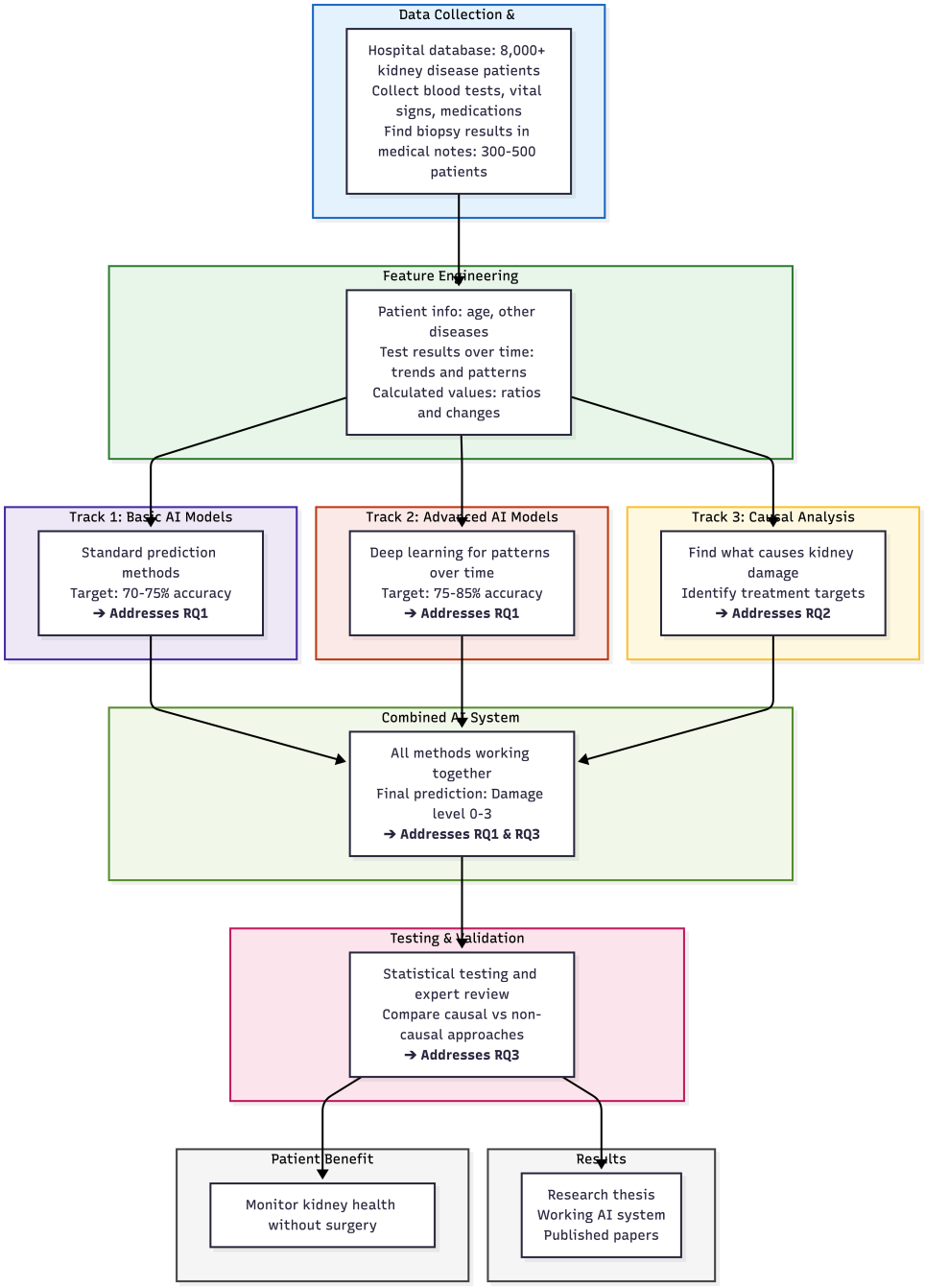


Figure 1: Overall Study Diagram (Virtual Kidney Biopsy Research Workflow)

Complete research methodology showing data collection from hospital database (8,000+ patients), feature engineering, three parallel AI modeling tracks (basic AI, advanced AI, and causal analysis), system integration, validation, and final outputs. Each track addresses specific research questions: Track 1 and 2 address RQ1 (prediction accuracy), Track 3 addresses RQ2 (causal relationships), and the validation phase addresses RQ3 (causal vs non causal comparison).

8. Timeline

Month	Activities
1	Data preparation: Download hospital database, extract kidney disease patients, find biopsy results in medical notes using text analysis, clean data
2	Build standard AI models: Test multiple prediction methods, select best performers, optimize settings
3	Build advanced models and apply causal analysis: Develop time pattern models, run causal algorithms, identify cause effect relationships
4	System integration: Combine all AI approaches into single system, test treatment simulations
4	Testing and validation: Statistical testing, expert review, compare causal vs non causal methods, document findings
5	Thesis writing: Complete all chapters, prepare research papers, create defense presentation

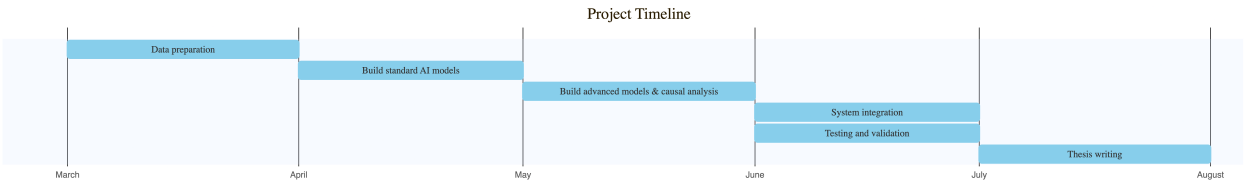


Figure 2: Project Implementation Timeline

This Gantt chart outlines the chronological progression of the research activities from March through August. The workflow demonstrates a sequential approach for the foundational and analytical phases, transitioning into parallel tasks during the evaluation stage.

9. Hardware/Software Resources Required

9.1 Hardware

- Personal laptop (sufficient for data processing and standard AI models)
- Google Colab (free cloud computing with GPU for advanced AI training)

9.2 Software

- Python 3.11: Primary programming language
- PostgreSQL: Database to store hospital data
- Pandas, NumPy: Data processing tools
- spaCy, NLTK: Text analysis for finding biopsy results
- scikit learn, XGBoost: Standard AI methods
- TensorFlow or PyTorch: Advanced AI methods
- DoWhy, CausalNex: Causal analysis tools
- Matplotlib, Seaborn: Data visualization

9.3 Data Access and Ethical Compliance

- **Data Source:** MIMIC-IV v3.1 (Medical Information Mart for Intensive Care, Version 4)
- **Data Provider:** PhysioNet (MIT Laboratory for Computational Physiology)
- **Access Status:** Approved (April 2026)
- **Ethical Requirements Met:**

    1. **CITI Program Training:** Completed
        - Certificate ID: 76046135
        - Course: Data or Specimens Only Research
        - Completion Date: March 22, 2026
        - Expiration Date: March 22, 2029
    2. **Data Use Agreement:** Signed with PhysioNet
    3. **Research Purpose:** Noninvasive kidney tissue assessment using deidentified patient data
    4. **Privacy Compliance:** All data is deidentified according to HIPAA Safe Harbor standards; no patient reidentification will be attempted
    5. **Institutional Approval:** Research conducted under supervision at Informatics Institute of Technology